

Chapter 1, Problem 25P

Problem

A 1.2-kW toaster takes roughly 4 minutes to heat four slices of bread. Find the cost of operating the toaster twice per day for 2 weeks (14 days). Assume energy costs 9 cents/kWh.

Step-by-step solution

Step 1 of 3

The power of a toaster is **1.2 kW**. It works for 4 minutes twice a day for two weeks. Therefore it works for

$$\begin{aligned} t &= 2 \times \frac{(14)(4)}{60} \\ &= \frac{112}{60} \text{ h} \end{aligned}$$

Step 2 of 3

The energy consumed by the toaster is,

$$\begin{aligned} w &= pt \\ &= (1.2) \left(\frac{112}{60} \right) \text{ kWh} \\ &= 2.24 \text{ kWh} \end{aligned}$$

Step 3 of 3

The cost per kWh is 9 cents.

The total cost of operating the toaster is,

$$\begin{aligned} \text{total cost} &= 9(2.24) \\ &= 20.16 \text{ cents} \end{aligned}$$

Therefore, the cost of operating the toaster 4 minutes twice per day for 2 weeks is

$$\boxed{20.16 \text{ cents}}.$$